Emotions in Physiological Signals

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Agenda

1. Physiological Signals
2. Heart Rate and Emotions
3. Skin Conductance and Emotions
4. EEG and Emotions

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Physiological Signals

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Emotion and Physiology

- Originate from the activity of the Autonomic Nervous System (ANS) which cannot be triggered by any conscious or intentional control.
- Taps the pure, unaltered emotion in contrast to modalities like facial expressions which can be faked.
- It also does not require the user's attention, which is important especially when assessing emotions among the individuals who are attention deficit.
- Recent advancements in the wearable technologies has shown a vital potential for hassle free acquisition of the physiological signals in an unintrusive manner.

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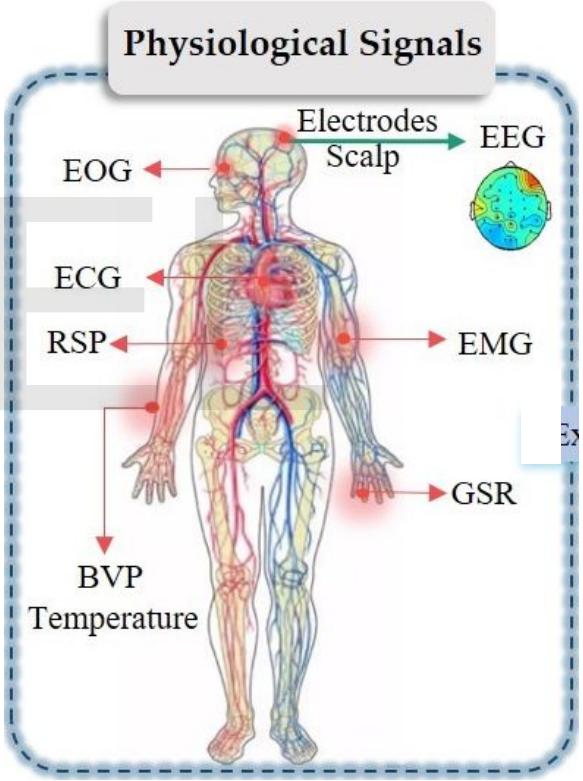
Emotion and Physiology

- Common measures include
  - Blood Presure (BP),
  - Electrocardiogram (ECG),
  - Electroencephalogram (EEG),
  - Electromyogram (EMG),
  - Galvanic skin response (GSR),
  - Heart Rate (HR),
  - Respiration (RSP),
  - Temperature (T) etc.



Source - https://encyclopedia.mdpi.cn/FgYJ7tjA2_k2FzvlajZoBb_OVmZh

Heart Rate and Emotions

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Heart Rate

- The activity of the heart also interacts with the brain, and ultimately changes how we feel.
  - Trying to recall something in front of an audience!

- Heart activity is closely linked to arousal, making it a great tool for helping to understand mental states in more detail.

- Indicator of overall physical activation and effort.

- Have been also reported as indicators of
  - Fear
  - Panic
  - Anger
  - Appreciation etc.

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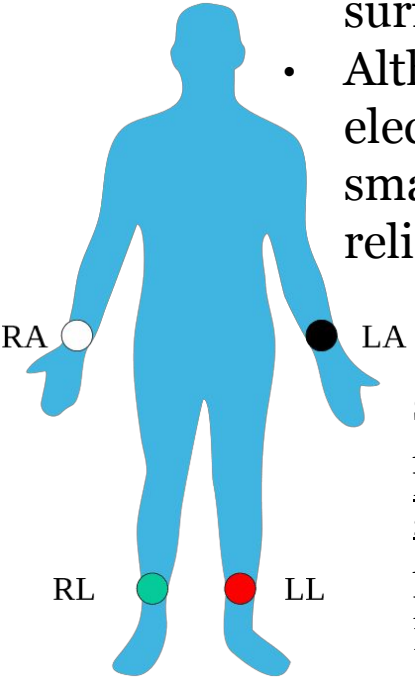
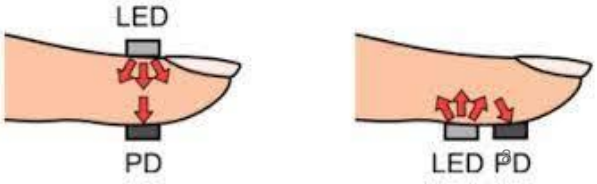
Heart Rate Measurement

- Electrocardiography (ECG, EKG): By monitoring the electrical changes on the surface of the skin.
- Although the amount of electricity is in fact very small, it can be picked up reliably (in microvolts or uV).



Source:
Left: https://media.imotions.com/images/20190411124422/Limb_leads.svg_.png
Right: Tamura, Toshiyo, et al. "Wearable photoplethysmographic sensors—past and present." Electronics 3.2 (2014): 282-302.

- Photo-Plethysmography (PPG): By measuring the pulse signals at various locations on the body, attached to the fingertip, the ear lobe or other capillary tissue.
- PPG clips use dry sensors and can be attached much quicker compared to ECG setups.
- Relatively easy to use, and a little bit less bothersome for participants.



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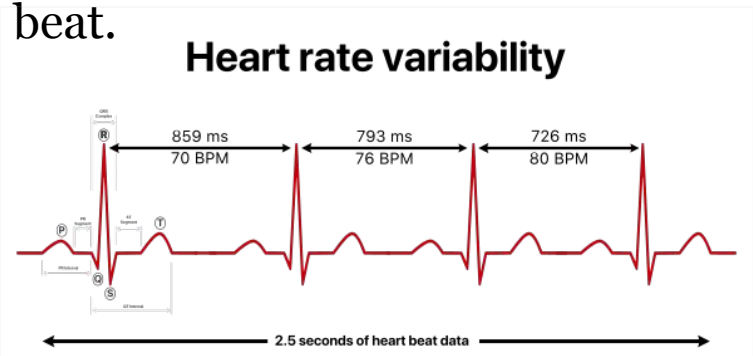
Cardiac Parameters

- Heart Rate (HR). Frequency of a complete heartbeat from its generation to the beginning of the next beat within a specific time window.
  - It is typically expressed as bpm.
  - An increased HR typically reflects increased arousal.

- Inter-Beat Interval (IBI): Time interval between individual beats of the heart, generally measured in units of milliseconds (ms).

- Heart Rate Variability (HRV): HRV expresses the natural variation of IBI values from beat to beat.



Source: Heart rate variability. (2022, September 12). In *Wikipedia*. https://en.wikipedia.org/wiki/Heart_rate_variability

Heart Rate and HRV

- HRV is closely related to emotional arousal and emotional regulation.

- It has been found to decrease under conditions of acute time pressure and emotional stress (meaning that the heartbeat is more consistent).

- Low HRV -> aroused, stress from exercise, psychological events, or other internal or external stressors.

- Higher HRV -> relaxed, body has a strong ability to tolerate stress or is strongly recovering from prior accumulated stress.

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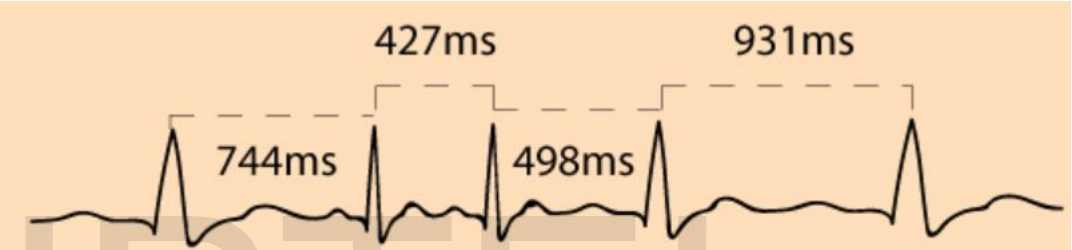
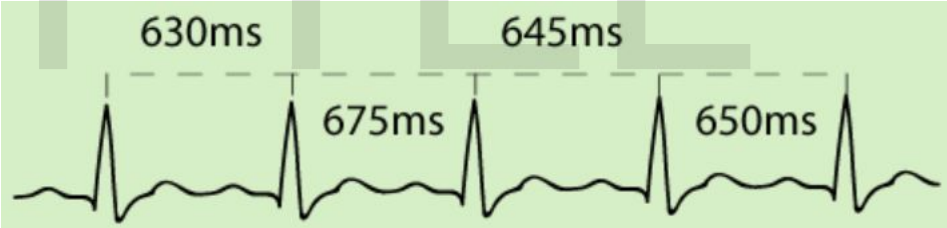
HRV Example

- Condition A



- Average_RR$_A$ = 650 ms

- Condition B



- Average_RR$_B$ = 650 ms

Source: https://imotions.com/blog/heart-rate-variability/

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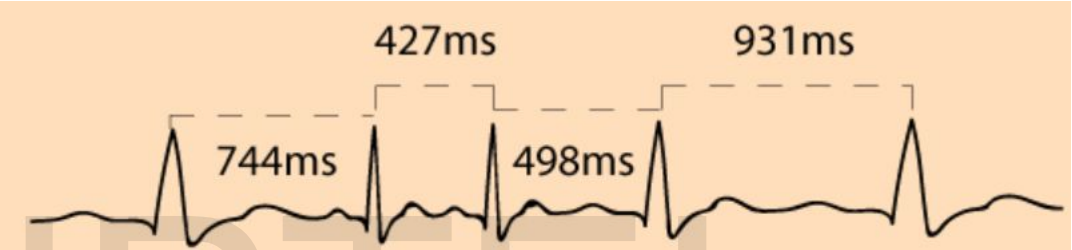
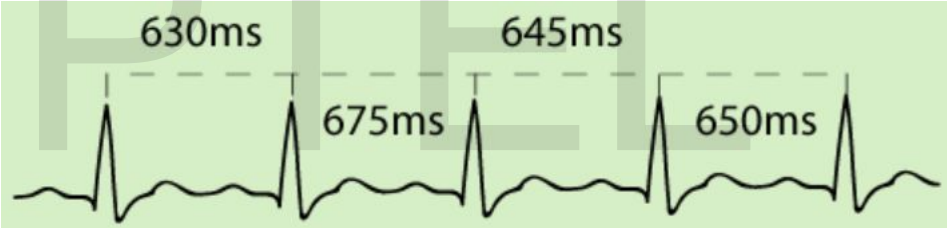
HRV Example

- Condition A



- Condition B



- Root Mean Square of Successive Differences = sqrt [mean{$(RR_1 - RR_2)^2 + (RR_2 - RR_3)^2 + \ldots$}]
- $mean_RMSSS_A = 313$ ms
- $mean_RMSSS_B = 31$ ms

Source: https://imotions.com/blog/heart-rate-variability/

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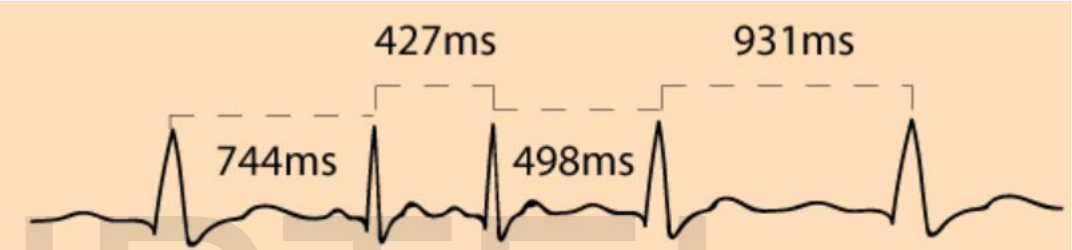
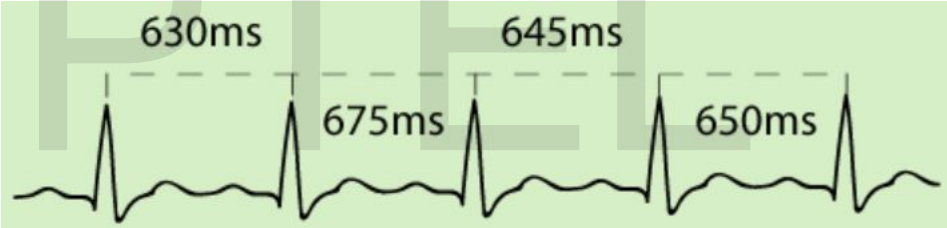
HRV Example

- Condition A



- Condition B



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- Average_RR$_A$= 650 ms | Average_RR$_B$= 650 ms
  - No difference between both the conditions when compared across beats per minutes or RR interval.
- mean_RMSSS$_A$ = 313 ms | mean_RMSSS$_B$ = 31 ms
  - Condition B is more physiologically arousing than condition A.

Source: https://imotions.com/blog/heart-rate-variability/

Other factors

- ECG can give a much more accurate instantaneous heart rate.
- Assumption for PPG based measurement: Heart rate time-series is stationary.
  - Longer time windows will give more accurate estimates.
  - 5 minutes (60 BPM) = 300 samples.

- Emotion is not the only factor that affects heart rate and heart rate variability.
  - Age
    - As age increases, the HRV decreases.
  - Posture
  - Level of physical conditioning
  - Breathing frequency etc.

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Limitations

- Data solely based on ECG or PPG data can't tell us whether the arousal was due to positive or negative stimulus content.
  - Both positive and negative stimuli can result in an increase in arousal triggering changes in heart rate.

- In other words: While ECG/PPG are ideal measures to track emotional arousal, they are not able to reveal emotional valence.

- Hence, usually these sensors are combined with other data sources such as facial expression analysis, EEG, and eye tracking.

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Skin Conductance and Emotions

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Skin Conductance: Emotional sweating

- Whenever we are emotionally aroused, the electrical conductivity of our skin subtly changes in response to both positive and negative stimuli's.
- Skin Conductance (SC), aka as *Electrodermal Activity (EDA)* or *Galvanic Skin Response (GSR)*, is a robust noninvasive way to measure this activation, caused by the amount of sweat in a person's sweat gland.
  - The sweating doesn't even need to be visible.
- GSR was first used by Carl Jung to identify "negative complexes" in word association tests.
- GSR is regarded as the most popular method for investigating human psychophysiological phenomena.
- Key component in lie detector tests.

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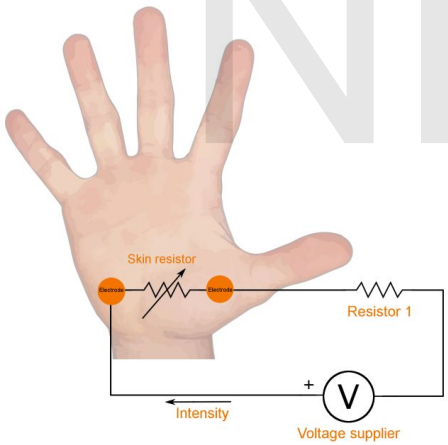
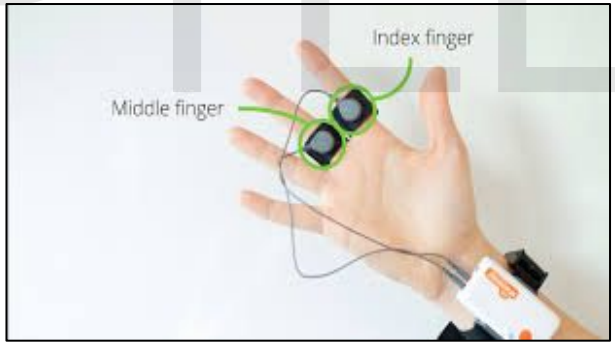
GSR Measurement

- GSR responses are extremely easy to measure as it can be measured anywhere on the body.
- The most emotionally reactive sweat glands are concentrated on the palms of the hands and the soles of the feet.





Left: GSR sensor schematics (Source: Tobii)
Middle: Shimmer GSR sensor. (Source: iMotions).
Right: E4 Wristband (Source: Empatica)

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GSR: Response Time

- Range between 2 and 20μS, and vary within a range between 1 and 3μS for different individuals.
- Typical rise time ~1 to 3 seconds and recovery time ~2 to 10 sec.
- Data is acquired with sampling rates between 1 – 10 Hz.

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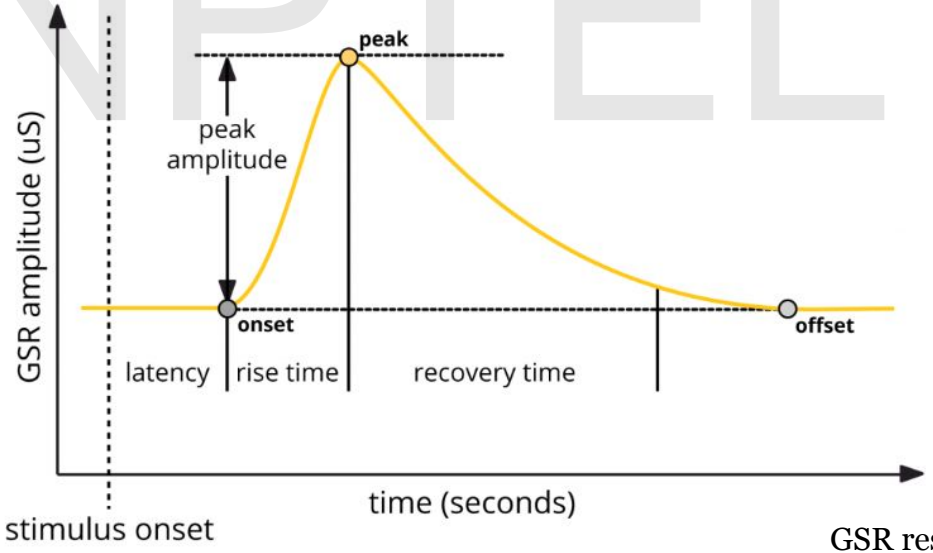
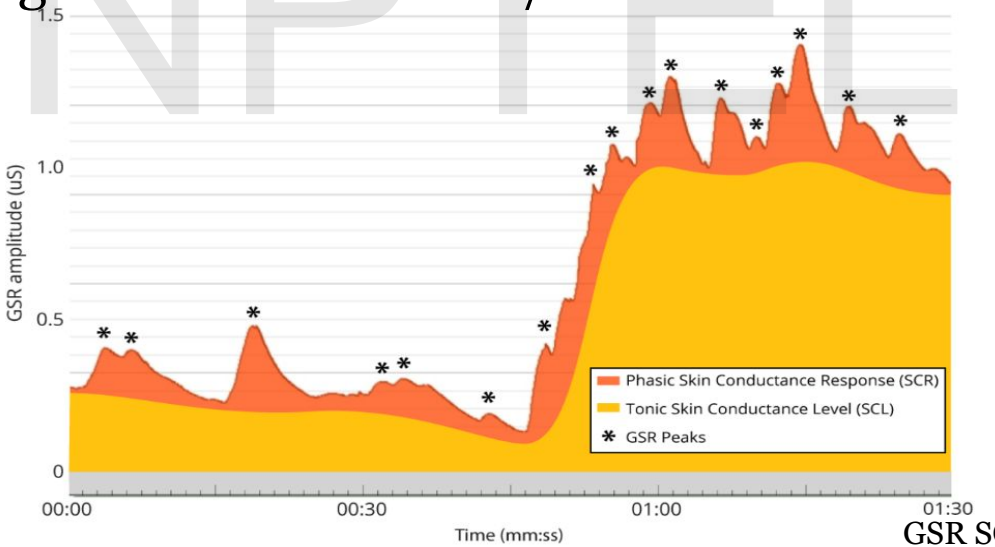


GSR response time (Source: iMotions)

GSR Signals: SCL and SCR

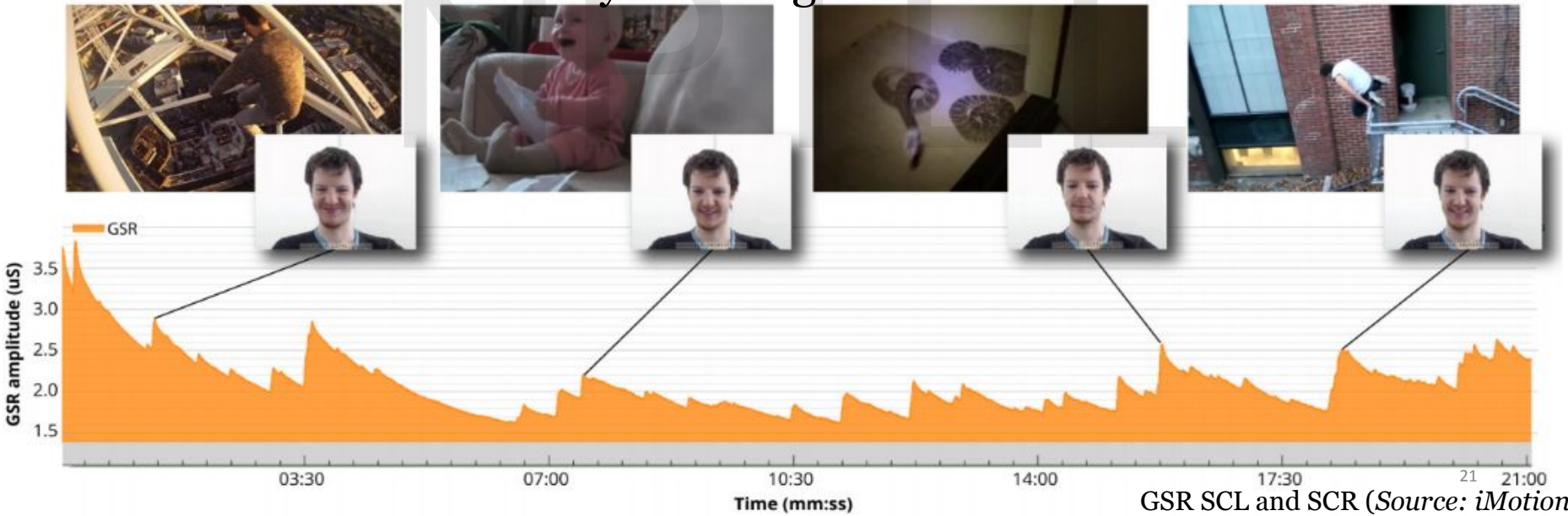
- For an emotional reaction to an external event (e.g., visual stimulus), the phasic or rapid component of the GSR signal, aka as skin conductance response (SCR), increases its amplitude temporarily.
- The tonic or slow component, aka as skin conductance level (SCL), also changes due to general arousal level/stress.



GSR SCL and SCR (*Source: iMotions*)

GSR Signals: SCL and SCR

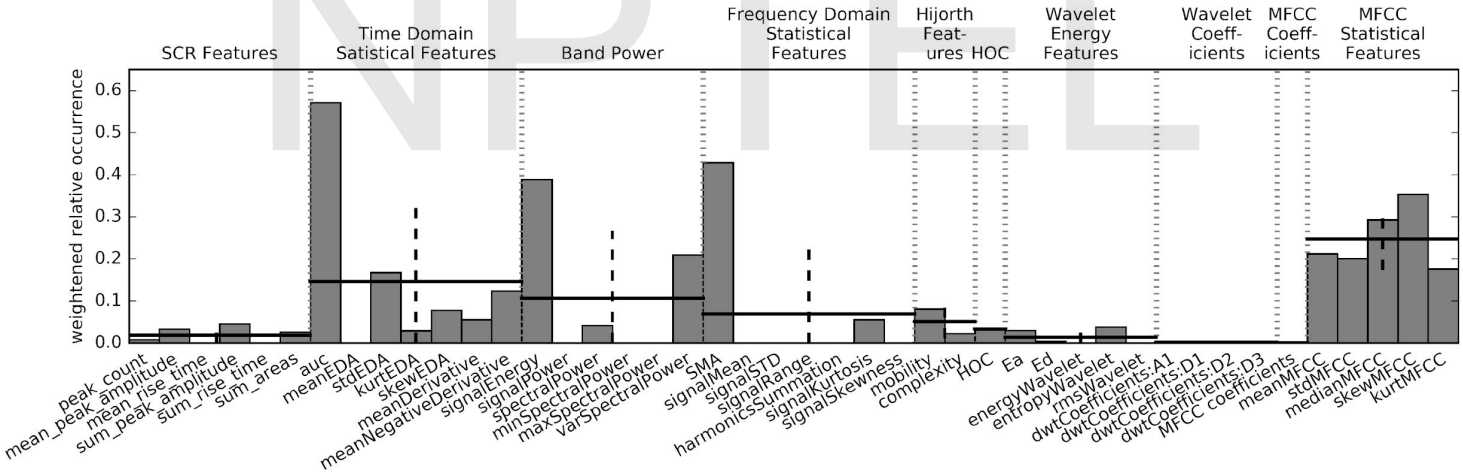
- To study event related SCRs (aka as ER-SCRs), it is important to display the stimulus for enough time (~10 seconds) and with similar inter-stimulus-interval durations.
- Which are the emotionally arousing scenes in the video?



GSR SCL and SCR (*Source: iMotions*)

Skin Conductance: Features

- Event related features
  - Mean Peak Amplitude, Sum Peak Amplitude, Peaks Count, Mean Rise Time etc.
- Band power [0.5-5 Hz] in frequency domain: Min, Max, Variance.
- Statistical MFCC features outperforms the commonly used SCR features



Shukla, Jainendra, et Al. "Feature extraction and selection for emotion recognition from electrodermal activity." IEEE Transactions on Affective Computing 12, no. 4 (2019): 857-869.

Limitations

- Annotations of emotions
  - Onset and the offset of the emotion is unknown in the wild.
- A central problem of analyzing the GSR data is how to separate SCR and SCL signals.
- Data solely based on GSR data can't tell us whether the arousal was due to positive or negative stimulus content.
  - Both positive and negative stimuli can result in an increase in arousal triggering GSR peaks.
- In other words: While GSR are ideal measures to track emotional arousal, they are not able to reveal emotional valence.
- Hence, usually GSR is also combined with other data sources such as facial expression analysis, EEG, and eye tracking.

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EEG and Emotions

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Electroencephalography (EEG)

- Whenever thousands of neurons fire in sync, they generate an electrical field which is strong enough to spread through tissue, bone, and skull.
  - Eventually, it can be measured on the head surface.
- EEG records the electrical activity generated by the brain via electrodes placed on the scalp surface.
- EEG reflects how the many neurons in the brain communicate with each other via electrical impulses, and how they are associated with cognitive processes such as drowsiness/alertness, wakeful relaxation, and approach or avoidance.
- It provides excellent time resolution, allowing detection of the activity within cortical areas-even at sub-second timescales

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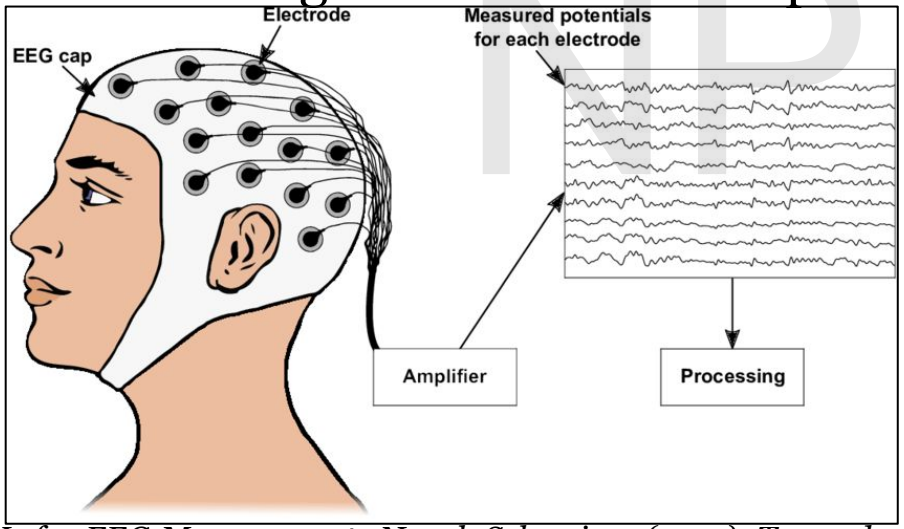
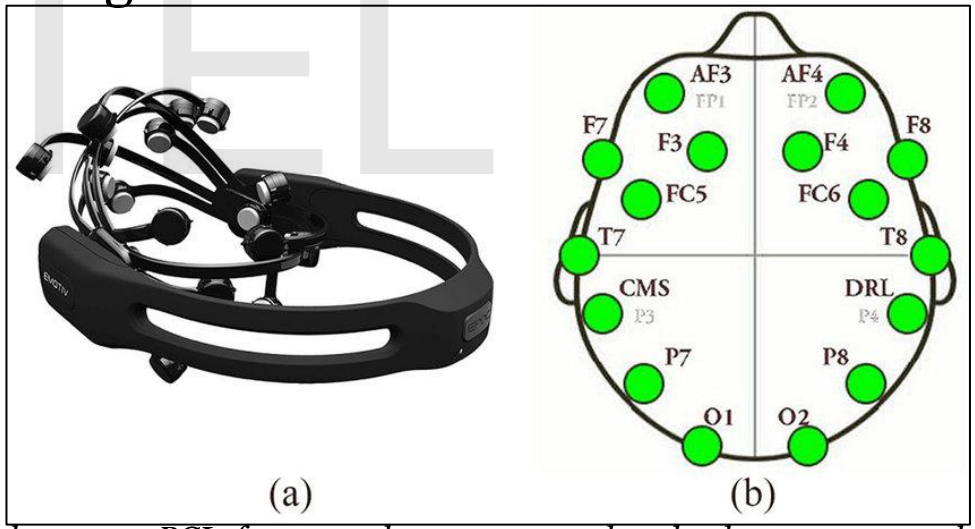
EEG Measurement

- One of the most non-invasive and accessible tools for analyzing brain activity.
- EEG electrodes record the EEG signals which is then amplified, digitized for storage and further data processing.

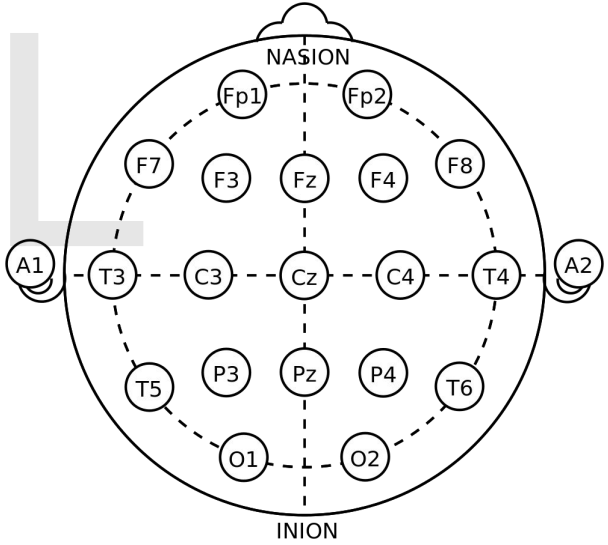




Left - EEG Measurement: *Nagel, Sebastian. (2019). Towards a home-use BCI: fast asynchronous control and robust non-control state detection. 10.15496/publikation-37739.* | *Right – (a)* Emotiv EPOC Headset *(b)* Electrode locations as per 10-20 system

EEG Electrode Labelling: 10–20 system

- Each electrode placement site has a letter to identify the lobe, or area of the brain it is reading from: pre-frontal (Fp), frontal (F), temporal (T), parietal (P), occipital (O), and central (C).
- A "Z" (zero) refers to an electrode placed on the midline sagittal plane of the skull, (FpZ, Fz, Cz, Oz) and is present mostly for reference/measurement points.
- Even-numbered electrodes (2,4,6,8) refer to electrode placement on the right side of the head, whereas odd numbers (1,3,5,7) refer to those on the left.



Electrode locations of International 10-20 system for EEG (Source: https://en.wikipedia.org/wiki/10%E2%80%9320_system_(EEG))

Frontal alpha asymmetry (FAA)

- The difference between right and left alpha activity over frontal regions.
  - Frequency power over left and right frontal regions (typically F3/F4 and F7/F8)
- FAA is used as a proxy for feelings of approach or avoidance.
- Beta (13-25 Hz), Gamma (>30 Hz) or Alpha (8-12 Hz) frequency bands are used.
- Inverse relationship exists between alpha band power and cortical activity
  - More brain activity means less alpha power and vis a versa.
- Relatively increased left-frontal activity may serve as an index of approach motivation or related emotion (e.g., anger and joy).
- In contrast, relatively increased right-frontal activity may serve as an index of withdrawal motivation or related emotion (e.g., disgust, fear, and sadness).

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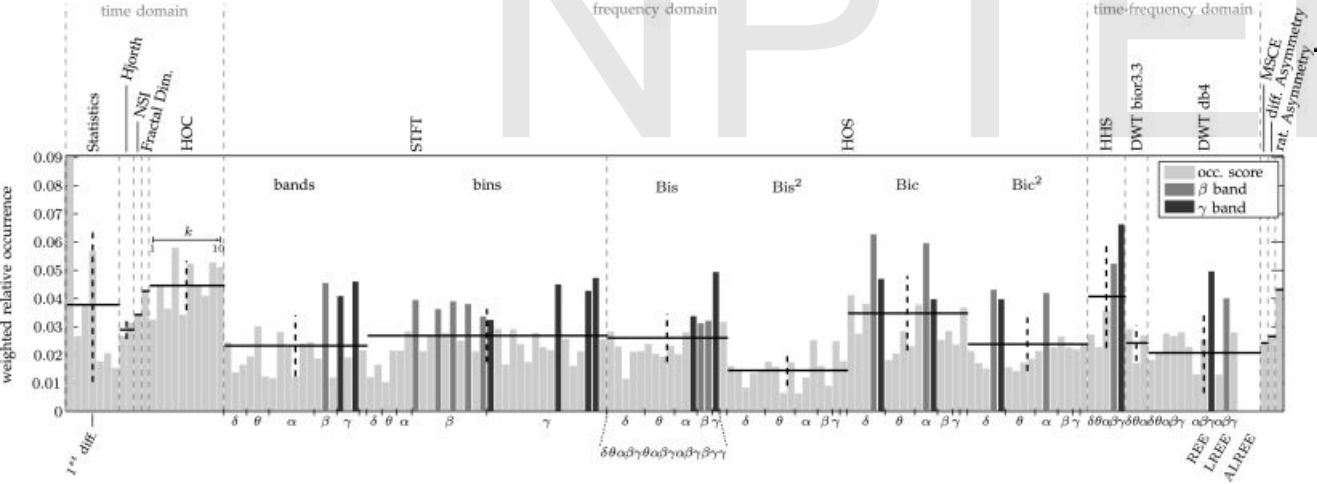
EEG Features

- Event-related potentials (ERPs) such as P300.
- Power features from frequency bands.
- Features like Higher Order Crossing (HOC) outperform commonly used power bands.

Bandwidth (Hz)	Frequency Band
1-4 Hz	δ
4-8 Hz	θ
8-10 Hz	slow α
8-12 Hz	α
12-30 Hz	β
30-64 Hz	γ



R. Jenke, A. Peer and M. Buss, "Feature Extraction and Selection for Emotion Recognition from EEG," in IEEE Transactions on Affective Computing, vol. 5, no. 3, pp. 327-339, 1 July-Sept. 2014, doi: 10.1109/TAFFC.2014.2339834.

Limitations

- A drawback for EEG is the poor spatial resolution.
  - As the electrodes measure electrical activity at the surface of the brain, it is difficult to know whether the signal was produced near the surface (in the cortex) or from a deeper.
- Intrusive in comparison to GSR and HR sensors.
- Some training is required to handle the EEG sensor.
- EEG data is almost always contaminated by such artifacts, i.e. the electrical signals detected along the scalp by an EEG, but are of non-cerebral origin.
- Usually EEG is also combined with other data sources such as facial expression analysis, and eye tracking.

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Conclusion

- Physiological Sensing is an important tool for AC community.
- It offers a continuous, discrete method for monitoring an individuals emotional state.
- HR, GSR and EEG are some of the most commonly used signals.
- Several Challenges:
  - Labelling: In the wild, the onset of an emotion is often unclear.
  - Subjective self-perception of emotion.
  - Many-to-one mapping of non-emotional and emotional influences on physiology.
  - Individual Variability
- With proper endeavor, physiological signals can enable affective computing in a private, personal and continuous way.

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